How-To Library

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Configuring an ODBC Connection for Google BigQuery

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Abstract

You can configure an ODBC connection to connect to Google BigQuery from Cloud Data Integration. This article explains how to configure an ODBC connection for Google BigQuery in Cloud Data Integration.

Supported Versions

- Cloud Data Integration Winter 2019

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Overview

Configure an ODBC connection to read data from and write data to Google BigQuery from Data Integration. You can use a Google BigQuery object as a source and as a target in synchronization tasks, mapping tasks, and mappings.

To establish an ODBC connection with Google BigQuery, perform the following tasks:

1. Create an ODBC data source.
2. Create an ODBC connection in Data Integration.

Before You Begin

Before you create an ODBC connection to connect to Google BigQuery, perform the following prerequisite tasks:

1. Create a Google account to access Google BigQuery. To create a Google service account, click the following URL: https://cloud.google.com/
2. On the **Credentials** page of the Google API Console, navigate to the APIs and auth section, and create a service account.
3. After you create the service account, download the .p12 key file.
4. Download the 2.1.19.1024 version of the Simba ODBC Driver with SQL Connector for BigQuery based on the operating system. To download the Simba ODBC Driver with SQL Connector for BigQuery, click the following URL: http://www.simba.com/product/google-bigquery-driver-with-sql-connector/
5. Install the Simba ODBC driver with SQL Connector for BigQuery.

Step 1. Create an ODBC Data Source

After you install the required components, you must create an ODBC data source for the Google BigQuery object that you want to connect to. A data source contains the connection credentials that you need to access the Google BigQuery object, such as client_email value, .p12 key file path, SQLDialect value, and Catalog name.

Create the ODBC data source on the machine from which you want to connect to Google BigQuery.

When you create an ODBC data source, you must also specify the driver to which the ODBC driver manager sends database calls. The Simba ODBC driver varies based on the operating systems.

Create an ODBC Data Source on Windows

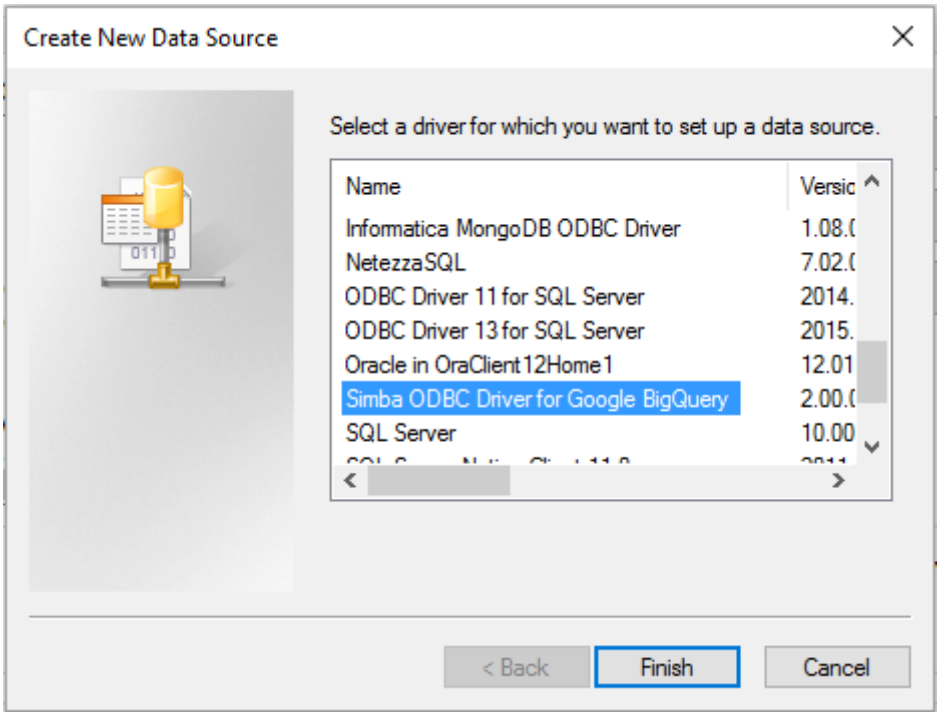
Use the ODBC Data Source Administrator to create an ODBC data source on Windows.

1. Click **Start** > **Control Panel**.
2. Click **Administrative Tools**.
3. Click **Data Sources (ODBC)**.

   The **ODBC Data Source Administrator** dialog box appears.
4. Click the **System DSN** tab.
5. Click **Add**.

   The **Create New Data Source** dialog box appears.
6. Select the Simba ODBC Driver for Google BigQuery to set up the data source.

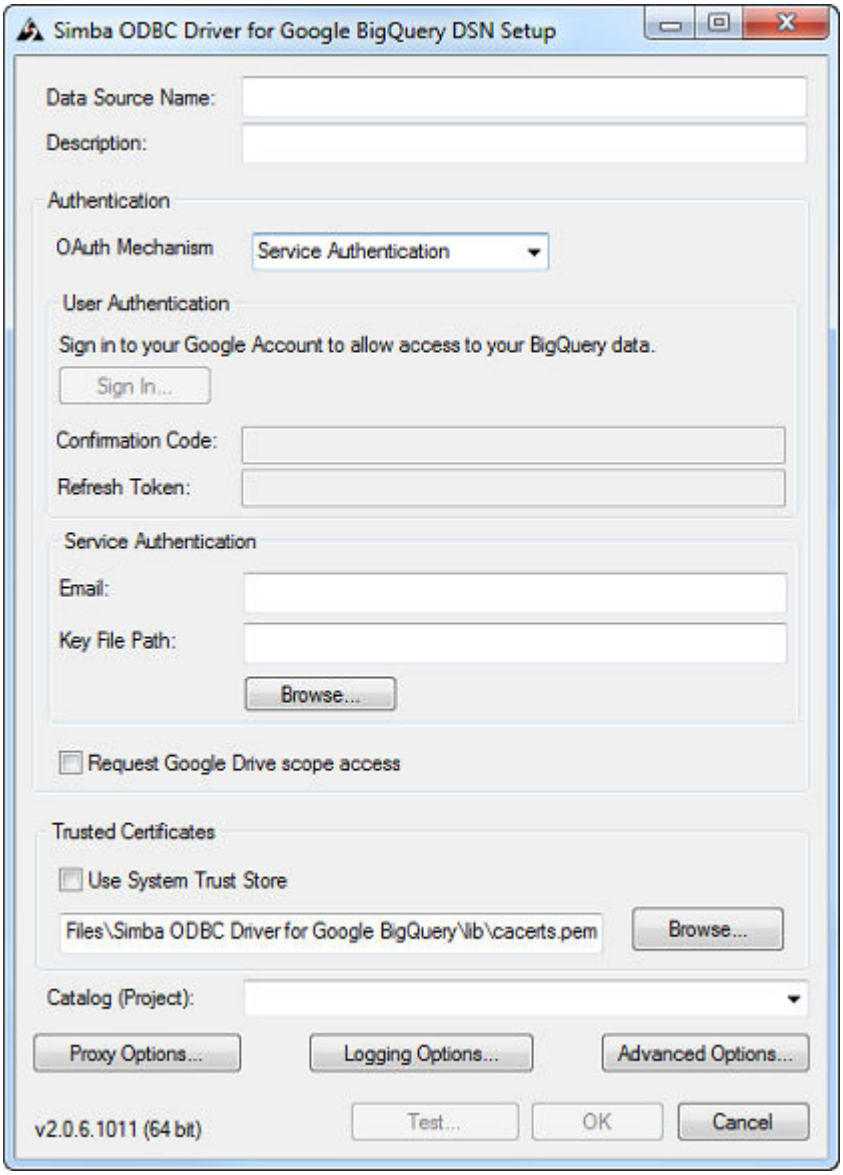
   The following image shows the Simba ODBC driver for Google BigQuery:



7. Click **Finish**.

   The **Simba ODBC Driver for Google BigQuery DSN Setup** dialog box appears.

The following image shows the connection configuration properties in the **Simba ODBC Driver for Google BigQuery DSN Setup** dialog box:



8. Configure the connection configuration properties.

   The following table describes the connection configuration properties that you must configure:

Property	Description
Data Source Name	Enter a name for the data source. The ODBC Driver for Google BigQuery uses the DSN to connect to the Google BigQuery.
Description	Enter a description.

Property	Description
OAuth Mechanism	The OAuth 2.0 authentication mechanism used to authenticate the driver. Select the **Service Authentication** option to authenticate the driver through a Google service account. If you select the **User Authentication** option, you will need to sign in to your Google service account. Click on **Sign In** and in the browser that opens, enter your credentials and then click **Sign In**. Click **Accept** to allow the Google BigQuery Client Tools to access Google BigQuery objects.
Confirmation Code	Code that Google provides when you click **Accept**. Copy and paste the code in the **Confirmation Code** field.
Refresh Token	The OAuth Mechanism populates the **Refresh Token** field when you paste the confirmation code.
Email	Specify the Google service account email id. This field is needed to authenticate the service account.
Key File Path	Enter the path to the .p12 key file that is used to authenticate the Google service account.
Request Google Drive scope access	Allows the driver to access Google Drive so that the driver can support federated tables that combine Google BigQuery data with data from Google Drive.
Trusted Certificates	Path of the `.pem` file. Use the trusted CA certificates from a specific `.pem` file or use the trusted CA certificates `.pem` file that is installed with the driver. **Note:** If you specify the Trusted Certificates .pem file path, you do not need to select the **Use System Trust Store** option.
Use System Trust Store	If you select the **Use System Trust Store** option, you do not need to specify the Trusted Certificates `.pem` file path.
Catalog (Project)	Name of the Google BigQuery project that the Simba ODBC Driver for Google BigQuery queries against. Also the project that is billed for queries that runs using the DSN.

9. Click **Proxy Options** to configure the connection to a data source through proxy server.
10. Click **Advanced Options** to configure the Simba ODBC Driver for Google BigQuery advanced properties.

    To read data from a Google BigQuery source, select **Legacy SQL** from the **SQLDialect** option in the **Advanced Options** dialog box.
    To write data to a Google BigQuery target, select **Standard SQL** from the **SQLDialect** option in the **Advanced Options** dialog box.
11. To enable logging, click **Logging Options**.
12. Click **Test** to verify the connection to Google BigQuery.
13. Click **OK** to close the **Simba ODBC Driver for Google BigQuery DSN Setup** dialog box.
14. Click **OK** to close the **ODBC Data Source Administrator** dialog box

    For more information about configuring an ODBC data source on Windows, see the Simba ODBC Driver with SQL Connector for Google BigQuery driver documentation.

Create an ODBC Data Source on UNIX

To create an ODBC data source on UNIX, configure the `odbc.ini` file.

1. Open the `odbc.ini` file.

2. Add the following driver parameters in the `odbc.ini` file:

```
[ODBC Data Sources]
Sample DSN=Simba ODBC Driver for Google BigQuery 64-bit
Driver=<Driver installation directory>/opt/simba/googlebigqueryodbc/lib/64/
libgooglebigqueryodbc_sb64.so
Catalog=<project_id>
SQLDialect=1
OAuthMech=0
Key File Path=<.p12 key file path>
```

   To read data from a Google BigQuery source, specify the value of **SQLDialect** as 0.
   To write data to a Google BigQuery target, specify the value of **SQLDialect** field as 1.

3. Set the ODBCINI environment variable on the UNIX machine to the location of the `odbc.ini` file.
4. Restart the Secure Agent.

   For more information about creating an ODBC data source on UNIX, see the Simba ODBC Driver with SQL Connector for Google BigQuery driver documentation.

Step 2. Create an ODBC Connection in Data Integration

To configure an ODBC connection for Google BigQuery, you must create an ODBC connection for Google BigQuery using the ODBC data source name in Data Integration.

1. In Administrator, select **Connections**.
2. Click **New Connection**.
3. Configure the following connection details:

Property	Description
Connection Name	Name of the connection.
Description	Description of the connection.
Type	Select **ODBC** from the list to create an ODBC connection for Google BigQuery.

4. Configure the following ODBC connection-specific properties:

Property	Description
Runtime Environment	Name of the runtime environment where you want to run the tasks.
User Name	Valid user name provided by Google to connect to Google BigQuery.
Password	Password for the user name.
Data Source Name	Name of the ODBC data source that you want to use to connect to Google BigQuery.
Schema	Name of the dataset present in Google BigQuery.
Code Page	Code page that the Secure Agent uses to read or write data.

Property	Description
ODBC Subtype	Type of database to which the ODBC driver connects. For Google BigQuery, select **Other**.
Driver Manager for Linux	Select **unixODBC 2.3.0** to read data from or write data to Google BigQuery.

5. To test the connection, click **Test Connection**.
6. Click **Save** to save the connection.

Rules and Guidelines

When you use an ODBC connection for Google BigQuery, consider the following rules and guidelines:

- When you read data from a Google BigQuery source, select **Legacy SQL** in the **SQLDialect** option in the **Advanced Options** dialog box when you create an ODBC data source for Google BigQuery.
- When you read data from a Google BigQuery source, you can add single tables as the source object for a Data Synchronization task or Mapping Configuration task.
- When you read data from a Google BigQuery source, you cannot add multiple tables in a source object for a Data Synchronization task or Mapping Configuration task.
- When you use a custom query as a source object in a Mapping Configuration task to read data from a Google BigQuery source, you can configure the Google BigQuery source to use Pre-SQL and Post-SQL commands.
- When you write to a Google BigQuery target, select **Standard SQL** in the **SQLDialect** option in the **Advanced Options** dialog box when you create an ODBC data source for Google BigQuery.
- When you write to a Google BigQuery target, the Secure Agent writes the first record from the source multiple times in the target and ignores the remaining records.
- When you write data to a Google BigQuery target, the **Truncate Target** option is not supported.
- When you read data from a Google BigQuery source, transform the data, and write the data to a Google BigQuery target, you cannot configure full pushdown optimization as the source and target connection are different.
- The **Legacy SQL** does not support DML statements for reading data from a Google BigQuery source.
- The Simba ODBC Driver with SQL Connector for Google BigQuery does not support key range partitioning when you use a Mapping Configuration task to read data from Google BigQuery sources.
- The **Legacy SQL** and **Standard SQL** supports custom query or saved query as a Google BigQuery source.

Author

Anupam Nayak

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